Real Estate Price Prediction

https://github.com/aiyshwary/real-estate-price-prediction

Python pandas NumPy scikit-learn Matplotlib Seaborn

OVERVIEW

A supervised regression case study that predicts house prices using the kc_house_data dataset, benchmarking multiple models and comparing performance with R2 and MSE metrics.

IMPACT

Benchmarked six regression models with bagging/boosting and hyperparameter tuning to identify the most accurate predictor for house prices.

KEY DETAILS

- Loaded and explored kc_house_data.csv, performing basic cleaning and feature analysis for regression modeling.
- Trained Linear Regression, KNN Regressor, Decision Tree, Random Forest, AdaBoost, and Gradient Boosting models.
- Compared model performance using R^2 and MSE to select the best-performing regressor.
- Applied bagging/boosting strategies and hyperparameter tuning to improve predictive accuracy.
- Documented the workflow in a Jupyter Notebook and included presentation slides for project reporting.